

Prime

Jojo encounter a prime problem. Jojo asked you to create a function called `checkPrime(int N)` to determine whether integer `N` is a prime number or not. A prime number is a natural number greater than 1 that cannot be formed by multiplying two smaller natural numbers.

Warning : Don't use `#include <cmath>`, `#include <math.h>` and built-in function, use recursive style.

You can do it recursively or iteratively. But it's "**strongly**" encouraged that you do it also in recursive style

Format Input

The first line consist of integer `T`. The next `T` line will consist of positive integer `N`. `N` will not exceed signed integer size.

Format Output

Output "Case #t: ", where `t` is the test case number starting from 1 along with the answer.

Constraints

$1 \leq T \leq 100$

$1 \leq N \leq 1000$

Sample Input	Sample Output
10	Case #1: YES
2	Case #2: YES
3	Case #3: NO
4	Case #4: YES
5	Case #5: NO
84	Case #6: NO
39	Case #7: NO
20	Case #8: YES
19	Case #9: NO
48	Case #10: NO
57	